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United States Patent	12404637
Kind Code	B2
Date of Patent	September 02, 2025
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Amplitude adjustment mechanism for a vibratory mechanism of a surface compaction machine

Abstract

An adjustment mechanism for a vibratory mechanism of a surface compaction machine, the adjustment mechanism includes a torque limiter coupled between the first eccentric shaft and the second eccentric shaft that prevents relative rotation between the shafts and a phase adjustment between the shafts when a net torque applied to the torque limiter is less than a locking torque threshold. Application of a net torque to the torque limiter that is greater than or equal to the locking torque threshold causes the first eccentric shaft to rotate with respect to the second eccentric shaft. An actuator subassembly selectively applies a linear force cause a first torque to be applied the first eccentric shaft sufficient to apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold to cause the first eccentric shaft to rotate with respect to the second eccentric shaft.

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Appl. No.:	17/908951
Filed (or PCT Filed):	March 04, 2020
PCT No.:	PCT/IB2020/051880
PCT Pub. No.:	WO2021/176250
PCT Pub. Date:	September 10, 2021

Prior Publication Data

Publication Classification

Int. Cl.: E01C19/28 (20060101); B06B1/16 (20060101); E02D3/074 (20060101)

U.S. Cl.:

CPC E01C19/286 (20130101); B06B1/164 (20130101); E02D3/074 (20130101);

Field of Classification Search

CPC: E01C (19/286); B06B (1/16); E02D (3/074)

USPC: 404/75; 404/117

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a 35 U.S.C. § 371 national stage application of PCT International Application No. PCT/IB2020/051880 filed on Mar. 4, 2020, the disclosure and content of which is incorporated by reference herein in its entirety.

FIELD

(2) Embodiments relate to a vibratory mechanism, and more particularly to an amplitude adjustment mechanism for a vibratory mechanism of a surface compaction machine.

BACKGROUND

(3) Surface compaction machines are used to compact a variety of substrates including soil, asphalt, or other materials. Surface compaction machines are provided with one or more compacting surfaces for this purpose. For example, a surface compaction machine, such as a roller compactor, may be provided with one or more cylindrical drums that provide compacting surfaces for compacting substrates.

(4) Roller compactors use the weight of the compactor applied through rolling drums to compress a surface of the substrate being rolled. In addition, one or more of the drums of some roller compactors may be vibrated by a vibration system to induce additional mechanical compaction of the substrate being rolled. The vibration system of these surface compaction machines can include an eccentric vibration system that includes an eccentric mass that is rotated to generate a vibration force which increases the compacting force exerted by the drum.

(5) These and other vibration systems may produce vibrations at different amplitudes by changing a combined center of mass of eccentric masses within the vibration system. These adjustments typically need to be performed manually, while the vibratory mechanism and surface compaction machine are not operating.

SUMMARY

(6) According to an embodiment, an adjustment mechanism for a vibratory mechanism of a surface compaction machine includes a screw coupled to a first eccentric shaft that is rotatable about an axis of rotation. The adjustment mechanism further includes a nut coupled to a second eccentric shaft that is rotatable about the axis of rotation, wherein the screw is disposed within the nut. The adjustment mechanism further includes a torque limiter coupled between the first eccentric shaft and the second eccentric shaft. The torque limiter prevents relative rotation between the first eccentric shaft and the second eccentric shaft and a phase adjustment between the first eccentric shaft and the second eccentric shaft when a net torque applied to the torque limiter is less than a locking torque threshold. Application of a net torque to the torque limiter that is greater than or equal to the locking torque threshold causes the first eccentric shaft to rotate with respect to the second eccentric shaft. The adjustment mechanism further includes an actuator subassembly coupled to the screw to selectively apply a first linear force to the screw in a linear direction parallel to the axis of rotation to cause the screw to apply a first torque to the first eccentric shaft. Application of the first torque to the first eccentric shaft causes the first eccentric shaft to apply the first torque to the first eccentric shaft sufficient to apply a net torque to the torque limiter that is

greater than or equal to the locking torque threshold to cause the first eccentric shaft to rotate with respect to the second eccentric shaft.

(7) According to another embodiment, a vibratory mechanism for a surface compaction machine includes a housing disposed within a compactor drum of the surface compaction machine. The vibratory mechanism further includes an eccentric shaft subassembly comprising a first eccentric shaft disposed within the housing, wherein the first eccentric shaft is rotatable about an axis of rotation, the eccentric shaft comprising a first eccentric mass having a first center of mass that is offset from the axis of rotation. The eccentric shaft subassembly further includes a second eccentric shaft disposed within the housing, wherein the second eccentric shaft is rotatable about the axis of rotation, the second eccentric shaft comprising a second eccentric mass having a second center of mass that is offset from the axis of rotation. The eccentric shaft subassembly further include a ball screw subassembly comprising a ball screw coupled to the first eccentric shaft, a ball nut coupled to the second eccentric shaft, wherein the ball screw is disposed within the ball nut, and a plurality of ball bearings disposed between the ball screw and the ball nut to reduce mechanical friction between the ball screw and the ball nut. The vibratory mechanism further includes a torque limiter coupled between the first eccentric shaft and the second eccentric shaft. The torque limiter prevents relative rotation between the first eccentric shaft and the second eccentric shaft and a phase adjustment between the first eccentric shaft and the second eccentric shaft when a net torque applied to the torque limiter is less than a locking torque threshold. Application of a net torque to the torque limiter that is greater than or equal to the locking torque threshold causes the first eccentric shaft to rotate with respect to the second eccentric shaft. The vibratory mechanism further includes an actuator subassembly coupled to the ball screw to selectively apply a first linear force to the ball screw in a linear direction parallel to the axis of rotation to cause the ball screw to apply a first torque to the torque limiter via the first eccentric shaft. The vibratory mechanism further includes a motor coupled to the second eccentric shaft to apply a second torque to the torque limiter via the second eccentric shaft. The second torque does not overcome the locking torque threshold, and the first torque and the second torque cause the net torque that is greater than or equal to the locking torque threshold to cause the first eccentric shaft to rotate with respect to the second eccentric shaft.

(8) According to another embodiment, a method of adjusting a vibratory mechanism of a surface compaction machine includes operating a motor to apply a first torque to a first eccentric shaft about an axis of rotation to rotate the first eccentric shaft. The first torque is less than a locking torque threshold of a torque limiter coupled to the first eccentric shaft. Rotating the first eccentric shaft causes concurrent rotation of a second eccentric shaft coupled to the torque limiter. The method further includes operating an actuator to selectively apply a second torque to the second eccentric shaft about the axis of rotation. The first torque and the second torque apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold of the torque limiter. Applying the first torque and the second torque causes the second eccentric shaft to rotate with respect to the first eccentric shaft.

(9) Other devices, methods, and systems according to embodiments will be or become apparent to one with skill in the art upon review of the following drawings and detailed description. It is intended that all such additional surface compaction machines, methods, and control systems be included within this description and protected by the accompanying claims. Moreover, it is intended that all embodiments disclosed herein can be implemented separately or combined in any way and/or combination.

Aspects

(10) According to an aspect, an adjustment mechanism for a vibratory mechanism of a surface compaction machine includes a screw coupled to a first eccentric shaft that is rotatable about an axis of rotation. The adjustment mechanism further includes a nut coupled to a second eccentric shaft that is rotatable about the axis of rotation, wherein the screw is disposed within the nut. The

adjustment mechanism further includes a torque limiter coupled between the first eccentric shaft and the second eccentric shaft. The torque limiter prevents relative rotation between the first eccentric shaft and the second eccentric shaft and a phase adjustment between the first eccentric shaft and the second eccentric shaft when a net torque applied to the torque limiter is less than a locking torque threshold. Application of a net torque to the torque limiter that is greater than or equal to the locking torque threshold causes the first eccentric shaft to rotate with respect to the second eccentric shaft. The adjustment mechanism further includes an actuator subassembly coupled to the screw to selectively apply a first linear force to the screw in a linear direction parallel to the axis of rotation to cause the screw to apply a first torque to the first eccentric shaft. Application of the first torque to the first eccentric shaft causes the first eccentric shaft to apply the first torque to the first eccentric shaft sufficient to apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold to cause the first eccentric shaft to rotate with respect to the second eccentric shaft.

(11) According to another aspect, the screw comprises a ball screw, and the nut comprises a ball nut. The adjustment mechanism further comprises a plurality of ball bearings disposed between the ball screw and the ball nut to reduce mechanical friction between the ball screw and the ball nut.

(12) According to another aspect, the torque limiter further comprises a ball detent mechanism to selectively lock the first eccentric shaft with respect to the second eccentric shaft in one of a plurality of rotational positions when the net torque applied to the torque limiter is less than the locking torque threshold.

(13) According to another aspect, the torque limiter further comprises a slip clutch mechanism to selectively lock the first eccentric shaft with respect to the second eccentric shaft when the net torque applied to the torque limiter is less than the locking torque threshold.

(14) According to another aspect, the adjustment mechanism further includes a sensor coupled to the torque limiter to measure a change in rotational position of the first eccentric shaft with respect to the second eccentric shaft.

(15) According to another aspect, the actuator subassembly further comprises a linear actuator, a screw hub coupled to the screw, and a lever coupled between the linear actuator and the screw hub. Actuation of the linear actuator causes the lever to apply the first linear force to the screw to apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold.

(16) According to another aspect, the screw hub comprises an outer hub pivotably coupled to the lever, and an inner hub rotatably coupled to the outer hub and movably coupled to the second eccentric shaft. The inner hub is movable with respect to the second eccentric shaft in the linear direction, and rotation of the second eccentric shaft causes rotation of the inner hub.

(17) According to another aspect, the adjustment mechanism further comprises a ball joint spherical bushing coupled between the inner hub and the screw. The inner hub is rotatable with respect to the screw, and application of the first linear force from the inner hub to the spherical bushing causes the ball joint to apply the first linear force to the screw.

(18) According to another aspect, a vibratory mechanism for a surface compaction machine includes a housing disposed within a compactor drum of the surface compaction machine. The vibratory mechanism further includes an eccentric shaft subassembly comprising a first eccentric shaft disposed within the housing, wherein the first eccentric shaft is rotatable about an axis of rotation, the eccentric shaft comprising a first eccentric mass having a first center of mass that is offset from the axis of rotation. The eccentric shaft subassembly further includes a second eccentric shaft disposed within the housing, wherein the second eccentric shaft is rotatable about the axis of rotation, the second eccentric shaft comprising a second eccentric mass having a second center of mass that is offset from the axis of rotation. The eccentric shaft subassembly further include a ball screw subassembly comprising a ball screw coupled to the first eccentric shaft, a ball nut coupled to the second eccentric shaft, wherein the ball screw is disposed within the ball nut, and a plurality of ball bearings disposed between the ball screw and the ball nut to reduce mechanical friction

between the ball screw and the ball nut. The vibratory mechanism further includes a torque limiter coupled between the first eccentric shaft and the second eccentric shaft. The torque limiter prevents relative rotation between the first eccentric shaft and the second eccentric shaft and a phase adjustment between the first eccentric shaft and the second eccentric shaft when a net torque applied to the torque limiter is less than a locking torque threshold. Application of a net torque to the torque limiter that is greater than or equal to the locking torque threshold causes the first eccentric shaft to rotate with respect to the second eccentric shaft. The vibratory mechanism further includes an actuator subassembly coupled to the ball screw to selectively apply a first linear force to the ball screw in a linear direction parallel to the axis of rotation to cause the ball screw to apply a first torque to the torque limiter via the first eccentric shaft. The vibratory mechanism further includes a motor coupled to the second eccentric shaft to apply a second torque to the torque limiter via the second eccentric shaft. The second torque does not overcome the locking torque threshold, and the first torque and the second torque cause the net torque that is greater than or equal to the locking torque threshold to cause the first eccentric shaft to rotate with respect to the second eccentric shaft.

(19) According to another aspect, the torque limiter further comprises a ball detent mechanism to selectively lock the first eccentric shaft with respect to the second eccentric shaft in one of a plurality of rotational positions when the net torque applied to the torque limiter is less than the locking torque threshold.

(20) According to another aspect, the torque limiter further comprises a slip clutch mechanism selectively lock the first eccentric shaft with respect to the second eccentric shaft when the net torque applied to the torque limiter is less than the locking torque threshold.

(21) According to another aspect, the first center of mass and the second center of mass produce a combined center of mass having an effective distance from the axis of rotation. Rotation of the first eccentric shaft with respect to the second eccentric shaft changes the effective distance of the combined center of mass from a first effective distance corresponding to a first vibratory amplitude to a second effective distance (84') corresponding to a second vibratory amplitude.

(22) According to another aspect, a sensor coupled to the torque limiter to measure a change in rotational position of the first eccentric shaft with respect to the second eccentric shaft.

(23) According to another aspect, the actuator subassembly further comprises a linear actuator coupled to the housing, a ball screw hub coupled to the ball screw, and a lever coupled between the linear actuator and the ball screw hub. Actuation of the linear actuator causes the lever to apply the first linear force to the ball screw in the linear direction to apply the first torque to the torque limiter via the first eccentric shaft to apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold.

(24) According to another aspect, the ball screw hub comprises an outer hub pivotably coupled to the lever, and an inner hub rotatably coupled to the outer hub and movably coupled to the second eccentric shaft. The inner hub is movable with respect to the second eccentric shaft in the linear direction, and wherein rotation of the second eccentric shaft causes rotation of the inner hub.

(25) According to another aspect, the vibratory mechanism further includes a ball joint coupled between the inner hub and the ball screw. The inner hub is rotatable with respect to the ball screw, and application of the first linear force from the inner hub to the ball joint causes the ball joint to apply the first linear force to the ball screw.

(26) According to another aspect, the vibratory mechanism further includes a spline mechanism coupled between the ball screw and the first eccentric shaft, wherein the spline mechanism permits linear movement of the ball screw with respect to the first eccentric shaft in the linear direction, and wherein the spline mechanism prevents rotation of the ball screw with respect to the first eccentric shaft.

(27) According to another aspect, a method of adjusting a vibratory mechanism of a surface compaction machine includes operating a motor to apply a first torque to a first eccentric shaft

about an axis of rotation to rotate the first eccentric shaft. The first torque is less than a locking torque threshold of a torque limiter coupled to the first eccentric shaft. Rotating the first eccentric shaft causes concurrent rotation of a second eccentric shaft coupled to the torque limiter. The method further includes operating an actuator to selectively apply a second torque to the second eccentric shaft about the axis of rotation. The first torque and the second torque apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold of the torque limiter. Applying the first torque and the second torque causes the second eccentric shaft to rotate with respect to the first eccentric shaft.

(28) According to another aspect, a first center of mass of the first eccentric shaft and a second center of mass of the second eccentric shaft produce a combined center of mass having an effective distance from the axis of rotation. Rotation of the first eccentric shaft with respect to the second eccentric shaft changes the effective distance of the combined center of mass from a first effective distance corresponding to a first vibratory amplitude to a second effective distance corresponding to a second vibratory amplitude.

(29) According to another aspect, the method further includes further operating the actuator to selectively remove the second torque from the second eccentric shaft about the axis of rotation to cause concurrent rotation of the second eccentric and the first eccentric shaft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this application, illustrate certain non-limiting embodiments of inventive concepts. In the drawings:

(2) FIG. 1 is an isometric view of a vibratory mechanism in a drum of a surface compaction machine having an adjustment mechanism for selectively modifying a vibratory amplitude of the vibratory mechanism, according to some embodiments;

(3) FIG. 2 is an isometric view of the vibratory mechanism of FIG. 1 illustrating components of the vibratory mechanism and adjustment mechanism, according to some embodiments;

(4) FIG. 3 is a cross-sectional view of the vibratory mechanism of FIGS. 1 and 2 illustrating additional components of the vibratory mechanism and adjustment mechanism, according to some embodiments;

(5) FIG. 4 is a detailed cross-sectional view of the adjustment mechanism of FIGS. 1-3 illustrating additional components of the adjustment mechanism, according to some embodiments;

(6) FIGS. 5A and 5B are a side view and cross-sectional view of the vibratory mechanism of FIGS. 1-4, wherein the adjustment mechanism causes relative rotation of the eccentric masses of the vibratory mechanism to correspond to a first vibratory amplitude, according to some embodiments;

(7) FIGS. 6A and 6B are a side view and cross-sectional view of the vibratory mechanism of FIGS. 1-5B, wherein the adjustment mechanism causes relative rotation of the eccentric masses of the vibratory mechanism to correspond to a second vibratory amplitude, according to some embodiments; and

(8) FIG. 7 is a flowchart of operations for a method of adjusting the vibratory mechanism of FIGS. 1-6B, according to some embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

(9) FIG. 1 is an isometric view of a vibratory mechanism **18** in a drum **14** of a surface compaction machine **10**. The surface compaction machine **10**, which may also be referred to herein as a vibratory compaction machine or a roller compactor, includes a body chassis structure **12**, and one or more rotatable drums **14** coupled to the body chassis structure **12** using a yoke **16**. The drum **14** may be driven by a drive motor (not shown) to propel the surface compaction machine **10**. In this

example, the cylindrical drum **14** is used to compact an underlying substrate, such as asphalt, gravel, soil, etc. Those of ordinary skill in the art will appreciate, however, that other types of surface compaction machines are contemplated, such as a surface compaction machine with multiple drums, for example, or other types of surface compaction machines and other equipment that utilize directional vibration energy.

(10) A vibratory mechanism **18** that generates vibration energy is mounted within the drum **14**. In this example, as discussed in greater detail below, the vibratory mechanism **18** is an eccentric vibration system having a drive motor **24** that rotates eccentric masses **20**, **22** to generate vibration energy, which causes the drum **14** to vibrate against the substrate to aid in compacting the substrate. Other types of vibration systems may be used within the drum **14** and/or at other locations of the surface compaction machine **10**, as well.

(11) Referring now to FIG. 2, an isometric view of the vibratory mechanism **18** of FIG. 1 illustrates additional components of the vibratory mechanism **18** and adjustment mechanism **26**, according to some embodiments. The vibratory mechanism **18** includes a pair of hubs **28** that may be coupled to the drum **14**, body chassis structure **12**, and/or other structure of the surface compaction machine **10** to secure the vibratory mechanism **18** within the drum **14**.

(12) The eccentric masses **20**, **22** are rotatably mounted between the hubs **28** via respective outer and inner eccentric shafts **46**, **48** (See FIG. 3) that rotate about a common axis of rotation. Each eccentric mass **20**, **22** has a center of mass that is offset from the axis of rotation. Based on the relative rotational positions of the eccentric masses **20**, **22** with respect to each other, the centers of mass of the eccentric masses **20**, **22** produce an effective center of mass that is an effective distance from the axis of rotation.

(13) In this embodiment, the drive motor **24** rotates the eccentric masses **20**, **22** about the axis of rotation at a common rotational speed to produce vibration energy at a particular frequency (based on the rotational speed) and amplitude (based on the effective distance of the effective center of mass of the eccentric masses **20**, **22**). Those of ordinary skill in the art will appreciate that it is desirable to selectively produce vibration energy at different amplitudes and/or frequencies. The frequency of the vibration energy can be selectively adjusted by varying the rotational speed of the drive motor **24**. As will be discussed in greater detail below, the amplitude of the vibration energy can be selectively adjusted by operating an adjustment mechanism **26** to vary the relative rotational positions of the eccentric masses **20**, **22** to modify an effective center of mass of the eccentric masses **20**, **22** with respect to an axis of rotation of the eccentric masses **20**, **22**.

(14) As shown in FIG. 2, the adjustment mechanism **26** includes a housing **30** that is fixed with respect to the drive motor **24** and that supports an actuator subassembly **32**. A lever **34** is coupled between the actuator subassembly **32** and a ball joint **36** that is fixed to the housing **30**. The lever is also pivotally connected to an outer hub **38** via a stone **40** and bushing **42** connection. Actuating the actuator subassembly **32** causes a linear actuator shaft **35** to pivot the lever about the ball joint **36**, which causes the outer hub **38** to move in a linear direction parallel to the axis of rotation of the eccentric masses **20**, **22**. It should also be understood that other mechanisms may be used to apply the torque to cause the inner eccentric shaft **48** and outer eccentric shaft **46** to rotate with respect to each other. For example, in some embodiments, a linear actuator may selectively apply the linear force directly to the outer hub **38**, or an actuator may selectively apply a rotational force directly to the inner eccentric shaft **48** or outer eccentric shaft **46** to cause the relative rotation.

(15) As will be described below with respect to FIGS. 3 and 4, the linear movement of the outer hub **38** applies a torque to the inner eccentric shaft **48** to cause the inner eccentric shaft **48** and outer eccentric shaft **46** to rotate with respect to each other. As will be described in greater detail with respect to FIGS. 5A-6B, this relative rotation of the outer eccentric shaft **46** and inner eccentric shaft **48** changes the effective distance of the effective center of mass of the eccentric masses **20**, **22**, thereby changing the vibratory amplitude of the vibratory mechanism **18**.

(16) Referring now to FIGS. 3 and 4, cross-sectional views of the vibratory mechanism **18** of FIGS.

1 and **2** illustrate additional components of the vibratory mechanism **18** and adjustment mechanism **26**, according to some embodiments. During operation of the vibratory mechanism in this embodiment, the drive motor **24** drives a cardan shaft **44**, which in turn drives the outer eccentric shaft **46** to rotate the first eccentric mass **20**. The outer eccentric shaft **46** is coupled to the inner eccentric shaft **48** via a torque limiter **56** that has a locking torque threshold. Application of a net torque to the torque limiter that is less than the locking torque threshold prevents relative rotation between the outer eccentric shaft **46** and the inner eccentric shaft **48** and a phase adjustment between the outer eccentric shaft **46** and the inner eccentric shaft **48**. The torque applied by the drive motor to the torque limiter via the outer eccentric shaft **46** in this operation is less than the locking torque threshold of the torque limiter **56**, and the inner eccentric shaft **48** and outer eccentric shaft **46** rotate together. In this example, the inner eccentric shaft **48** and outer eccentric shaft **46** are supported within the hubs **28** by roller bearings **47**, which facilitate rotation of the inner eccentric shaft **48** and outer eccentric shaft **46** with respect to the hubs **28**.

(17) However, application of a net torque to the torque limiter **56** that is greater than or equal to the locking torque threshold causes the outer eccentric shaft **46** and inner eccentric shaft **48** to rotate with respect to each other to change the relative rotational positions of the eccentric masses **20**, **22**. In this regard, actuation of the actuator subassembly **32** causes the outer hub **38** to apply a linear force to a ball screw **52** coupled to the inner eccentric shaft **48**. The ball screw **52** is disposed within a ball nut **54** coupled to the outer eccentric shaft **46**, such that the linear force applied to the ball screw **52** causes the ball screw **52** to apply an additional torque to the torque limiter **56** via the inner eccentric shaft **48**. The additional torque causes the net torque applied to the torque limiter **56** to overcome the locking torque threshold, thereby causing the inner eccentric shaft **48** to rotate with respect to the outer eccentric shaft **46**. In this example at least two needle bearings **50** are disposed between the inner eccentric shaft **48** and outer eccentric shaft **46** to facilitate rotation of the inner eccentric shaft **48** and outer eccentric shaft **46** with respect to each other.

(18) A sensor **58** is coupled to the torque limiter **56** to detect rotation of the inner eccentric shaft **48** and outer eccentric shaft **46** with respect to each other. The sensor **58** may be used to control the actuator subassembly **32** to obtain a desired vibratory amplitude for the vibratory mechanism **18**.

(19) Referring now to FIG. **4**, a detailed cross-sectional view of the adjustment mechanism **26** of FIGS. **1-3** illustrates additional components of the adjustment mechanism **26**, according to some embodiments. The cardan shaft **44** is coupled to the outer eccentric shaft **46** via a plurality of rails **64**. A screw hub **60** is coupled to the rails **64** via bushings **66** such that the screw hub **60** is moveable with respect to the rails **64** in a linear direction parallel to the axis of rotation. The screw hub **60** is coupled to the outer hub **38** via a plurality of bearings **62** that transfer linear force between the outer hub **38** and screw hub **60** while allowing the screw hub **60** to rotate freely with respect to the outer hub **38**. A screw shaft **68** coupled to the ball screw **52** is coupled to the screw hub **60** via a spherical bushing **70**, which transfers linear force between the screw hub **60** and the ball screw **52** while allowing the screw hub **60** to rotate freely with respect to the screw shaft **68** and ball screw **52**. In some embodiments, a thrust bearing may be substituted for the spherical bushing **70**. The ball screw **52** is coupled to the inner eccentric shaft **48** via a spline connection **72**, which transfers torque between the ball screw **52** and the inner eccentric shaft **48** while allowing linear movement of the ball screw **52** with respect to the inner eccentric shaft **48**. In this embodiment, one or more flexible covers **74**, such as a rubber cover, may enclose components of the adjustment mechanism **26** while allowing linear movement of the screw hub **60** and outer hub **38** along the rails **64**.

(20) FIGS. **5A** and **5B** are a side view and cross-sectional view of the vibratory mechanism **18** of FIGS. **1-4**, wherein the actuator subassembly **32** causes relative rotation of the eccentric masses **20**, **22** of the vibratory mechanism **18** to correspond to a first vibratory amplitude, according to some embodiments. As shown by FIG. **5A**, the linear actuator shaft **35** is at a first position that causes the outer eccentric shaft **46** and the inner eccentric shaft **48** to rotate relative to each other to produce a

first phase angle $\alpha 1$. As shown by FIG. 5B, which is a cross sectional view of the vibratory mechanism **18** at line A-A at the first phase angle $\alpha 1$, the first center of mass **76** of the first eccentric mass **20** and the second center of mass **78** of the second eccentric mass **22** produce a first combined center of mass **82** having a first effective distance **84** from the axis of rotation corresponding to a first vibratory amplitude for the vibratory mechanism **18**.

(21) FIGS. 6A and 6B are a side view and cross-sectional view of the vibratory mechanism **18** of FIGS. 1-5B at a second phase angle $\alpha 2$. As shown by FIG. 6A, the linear actuator shaft **35** is moved to a second position that causes the outer eccentric shaft **46** and the inner eccentric shaft **48** to rotate relative to each other to produce the second phase angle $\alpha 2$. As shown by FIG. 6B, relative rotation of the first center of mass **76** of the first eccentric mass **20** and the second center of mass **78** of the second eccentric mass **22** produce a second combined center of mass **82'** having a second effective distance **84'** from the axis of rotation corresponding to a second vibratory amplitude for the vibratory mechanism **18**.

(22) FIG. 7 is a flowchart of operations **700** for a method of adjusting the vibratory mechanism **18** of FIGS. 1-6B, according to some embodiments. The operations **700** include operating a motor to apply a first torque to a first eccentric shaft about an axis of rotation to rotate the first eccentric shaft, wherein the first torque is less than a locking torque threshold of a torque limiter coupled to the first eccentric shaft, and wherein rotating the first eccentric shaft causes concurrent rotation of a second eccentric shaft coupled to the torque limiter (Block **702**). The operations **700** further include operating an actuator to selectively apply a second torque to the second eccentric shaft about the axis of rotation, wherein the first torque and the second torque apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold of the torque limiter, and wherein applying the first torque and the second torque causes the second eccentric shaft to rotate with respect to the first eccentric shaft (Block **704**).

(23) These and other embodiments may have several advantages. For example, using a torque limiter allows the amplitude of the vibratory mechanism to be dynamically adjusted during operation of the vibratory mechanism and surface compaction machine. In addition, using a torque limiter helps prevent unintentional rotation of the shafts with respect to each other during operation, and allows for secure locking of the shafts with respect to each other in a non-static environment that is subject to vibration and temperature fluctuations. The torque limiter also helps to reduce wear on the ball screw and linear actuator, and may allow for greater rotational precision. Another advantage is that the amplitude of the vibratory mechanism can be dynamically adjusted during operation.

(24) When an element is referred to as being “connected”, “coupled”, “responsive”, “mounted”, or variants thereof to another element, it can be directly connected, coupled, responsive, or mounted to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected”, “directly coupled”, “directly responsive”, “directly mounted” or variants thereof to another element, there are no intervening elements present. Like numbers refer to like elements throughout. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Well-known functions or constructions may not be described in detail for brevity and/or clarity. The term “and/or” and its abbreviation “/” include any and all combinations of one or more of the associated listed items.

(25) It will be understood that although the terms first, second, third, etc. may be used herein to describe various elements/operations, these elements/operations should not be limited by these terms. These terms are only used to distinguish one element/operation from another element/operation. Thus, a first element/operation in some embodiments could be termed a second element/operation in other embodiments without departing from the teachings of present inventive concepts. The same reference numerals or the same reference designators denote the same or similar elements throughout the specification.

(26) As used herein, the terms “comprise”, “comprising”, “comprises”, “include”, “including”, “includes”, “have”, “has”, “having”, or variants thereof are open-ended, and include one or more stated features, integers, elements, steps, components or functions but do not preclude the presence or addition of one or more other features, integers, elements, steps, components, functions or groups thereof. Furthermore, as used herein, the common abbreviation “e.g.”, which derives from the Latin phrase “*exempli gratia*,” may be used to introduce or specify a general example or examples of a previously mentioned item, and is not intended to be limiting of such item. The common abbreviation “i.e.”, which derives from the Latin phrase “*id est*,” may be used to specify a particular item from a more general recitation.

(27) Persons skilled in the art will recognize that certain elements of the above-described embodiments may variously be combined or eliminated to create further embodiments, and such further embodiments fall within the scope and teachings of inventive concepts. It will also be apparent to those of ordinary skill in the art that the above-described embodiments may be combined in whole or in part to create additional embodiments within the scope and teachings of inventive concepts. Thus, although specific embodiments of, and examples for, inventive concepts are described herein for illustrative purposes, various equivalent modifications are possible within the scope of inventive concepts, as those skilled in the relevant art will recognize. Accordingly, the scope of inventive concepts is determined from the appended claims and equivalents thereof.

Claims

1. An adjustment mechanism for a vibratory mechanism of a surface compaction machine, the adjustment mechanism comprising: a screw coupled to a first eccentric shaft that is rotatable about an axis of rotation; a nut coupled to a second eccentric shaft that is rotatable about the axis of rotation, wherein the screw is disposed within the nut; a torque limiter coupled between the first eccentric shaft and the second eccentric shaft, wherein the torque limiter prevents relative rotation between the first eccentric shaft and the second eccentric shaft and a phase adjustment between the first eccentric shaft and the second eccentric shaft when a net torque applied to the torque limiter is less than a locking torque threshold, and wherein application of a net torque to the torque limiter that is greater than or equal to the locking torque threshold causes the first eccentric shaft to rotate with respect to the second eccentric shaft; and an actuator subassembly coupled to the screw to selectively apply a first linear force to the screw in a linear direction parallel to the axis of rotation to cause the screw to apply a first torque to the first eccentric shaft, wherein application of the first torque to the first eccentric shaft causes the first eccentric shaft to apply the first torque to the first eccentric shaft sufficient to apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold to cause the first eccentric shaft to rotate with respect to the second eccentric shaft.

2. The adjustment mechanism of claim 1, wherein the screw comprises a ball screw, wherein the nut comprises a ball nut, and wherein the adjustment mechanism further comprises a plurality of ball bearings disposed between the ball screw and the ball nut to reduce mechanical friction between the ball screw and the ball nut.

3. The adjustment mechanism of claim 1, wherein the torque limiter further comprises a ball detent mechanism to selectively lock the first eccentric shaft with respect to the second eccentric shaft in one of a plurality of rotational positions when the net torque applied to the torque limiter is less than the locking torque threshold.

4. The adjustment mechanism of claim 1, wherein the torque limiter further comprises a slip clutch mechanism to selectively lock the first eccentric shaft with respect to the second eccentric shaft when the net torque applied to the torque limiter is less than the locking torque threshold.

5. The adjustment mechanism of claim 1, further comprising a sensor coupled to the torque limiter to measure a change in rotational position of the first eccentric shaft with respect to the second

eccentric shaft.

6. The adjustment mechanism of claim 1, wherein the actuator subassembly further comprises: a linear actuator; a screw hub coupled to the screw; and a lever coupled between the linear actuator and the screw hub, wherein actuation of the linear actuator causes the lever to apply the first linear force to the screw to apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold.

7. The adjustment mechanism of claim 6, wherein the screw hub comprises: an outer hub pivotably coupled to the lever; and an inner hub rotatably coupled to the outer hub and movably coupled to the second eccentric shaft, wherein the inner hub is movable with respect to the second eccentric shaft in the linear direction, and wherein rotation of the second eccentric shaft causes rotation of the inner hub.

8. The adjustment mechanism of claim 7, further comprising a ball joint spherical bushing coupled between the inner hub and the screw, wherein the inner hub is rotatable with respect to the screw, and wherein application of the first linear force from the inner hub to the spherical bushing causes the ball joint to apply the first linear force to the screw.

9. A vibratory mechanism for a surface compaction machine, the vibratory mechanism comprising: a housing disposed within a compactor drum of the surface compaction machine; an eccentric shaft subassembly comprising: a first eccentric shaft disposed within the housing, wherein the first eccentric shaft is rotatable about an axis of rotation, the eccentric shaft comprising a first eccentric mass having a first center of mass that is offset from the axis of rotation; and a second eccentric shaft disposed within the housing, wherein the second eccentric shaft is rotatable about the axis of rotation, the second eccentric shaft comprising a second eccentric mass having a second center of mass that is offset from the axis of rotation; a ball screw subassembly comprising: a ball screw coupled to the first eccentric shaft; a ball nut coupled to the second eccentric shaft, wherein the ball screw is disposed within the ball nut; and a plurality of ball bearings disposed between the ball screw and the ball nut to reduce mechanical friction between the ball screw and the ball nut; a torque limiter coupled between the first eccentric shaft and the second eccentric shaft, wherein the torque limiter prevents relative rotation between the first eccentric shaft and the second eccentric shaft and a phase adjustment between the first eccentric shaft and the second eccentric shaft when a net torque applied to the torque limiter is less than a locking torque threshold, and wherein application of a net torque to the torque limiter that is greater than or equal to the locking torque threshold causes the first eccentric shaft to rotate with respect to the second eccentric shaft; and an actuator subassembly coupled to the ball screw to selectively apply a first linear force to the ball screw in a linear direction parallel to the axis of rotation to cause the ball screw to apply a first torque to the torque limiter via the first eccentric shaft; and a motor coupled to the second eccentric shaft to apply a second torque to the torque limiter via the second eccentric shaft, wherein the second torque does not overcome the locking torque threshold, and wherein the first torque and the second torque cause the net torque that is greater than or equal to the locking torque threshold to cause the first eccentric shaft to rotate with respect to the second eccentric shaft.

10. The vibratory mechanism of claim 9, wherein the torque limiter further comprises a ball detent mechanism to selectively lock the first eccentric shaft with respect to the second eccentric shaft in one of a plurality of rotational positions when the net torque applied to the torque limiter is less than the locking torque threshold.

11. The vibratory mechanism of claim 9, wherein the torque limiter further comprises a slip clutch mechanism selectively lock the first eccentric shaft with respect to the second eccentric shaft when the net torque applied to the torque limiter is less than the locking torque threshold.

12. The vibratory mechanism of claim 9, wherein the first center of mass and the second center of mass produce a combined center of mass having an effective distance from the axis of rotation, and wherein rotation of the first eccentric shaft with respect to the second eccentric shaft changes the effective distance of the combined center of mass from a first effective distance corresponding to a

first vibratory amplitude to a second effective distance (84 ') corresponding to a second vibratory amplitude.

13. The vibratory mechanism of claim 9, further comprising a sensor coupled to the torque limiter to measure a change in rotational position of the first eccentric shaft with respect to the second eccentric shaft.

14. The vibratory mechanism of claim 9, wherein the actuator subassembly further comprises: a linear actuator coupled to the housing; a ball screw hub coupled to the ball screw; and a lever coupled between the linear actuator and the ball screw hub, and wherein actuation of the linear actuator causes the lever to apply the first linear force to the ball screw in the linear direction to apply the first torque to the torque limiter via the first eccentric shaft to apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold.

15. The vibratory mechanism of claim 14, wherein the ball screw hub comprises: an outer hub pivotably coupled to the lever; and an inner hub rotatably coupled to the outer hub and movably coupled to the second eccentric shaft, wherein the inner hub is movable with respect to the second eccentric shaft in the linear direction, and wherein rotation of the second eccentric shaft causes rotation of the inner hub.

16. The vibratory mechanism of claim 15, further comprising a ball joint coupled between the inner hub and the ball screw, wherein the inner hub is rotatable with respect to the ball screw, and wherein application of the first linear force from the inner hub to the ball joint causes the ball joint to apply the first linear force to the ball screw.

17. The vibratory mechanism of claim 9, further comprising: a spline mechanism coupled between the ball screw and the first eccentric shaft, wherein the spline mechanism permits linear movement of the ball screw with respect to the first eccentric shaft in the linear direction, and wherein the spline mechanism prevents rotation of the ball screw with respect to the first eccentric shaft.

18. A method of adjusting a vibratory mechanism of a surface compaction machine, the method comprising: operating a motor to apply a first torque to a first eccentric shaft about an axis of rotation to rotate the first eccentric shaft, wherein the first torque is less than a locking torque threshold of a torque limiter coupled to the first eccentric shaft, and wherein rotating the first eccentric shaft causes concurrent rotation of a second eccentric shaft coupled to the torque limiter; and operating an actuator to selectively apply a second torque to the second eccentric shaft about the axis of rotation, wherein the first torque and the second torque apply a net torque to the torque limiter that is greater than or equal to the locking torque threshold of the torque limiter, and wherein applying the first torque and the second torque causes the second eccentric shaft to rotate with respect to the first eccentric shaft.

19. The method of claim 18, wherein a first center of mass of the first eccentric shaft and a second center of mass of the second eccentric shaft produce a combined center of mass having an effective distance from the axis of rotation, and wherein rotation of the first eccentric shaft with respect to the second eccentric shaft changes the effective distance of the combined center of mass from a first effective distance corresponding to a first vibratory amplitude to a second effective distance corresponding to a second vibratory amplitude.

20. The method of claim 18, further operating the actuator to selectively remove the second torque from the second eccentric shaft about the axis of rotation to cause concurrent rotation of the second eccentric and the first eccentric shaft.
